

Def. Let $I \subseteq \mathbb{R}$ be an interval, and let $f: I \rightarrow [0, \infty)$ be a non-negative valued. We said that $\int_I f$ converges if f is Riemann-Integrable on any $[c, d] \subseteq I$ with

$$\sup \left\{ \int_c^d f \mid [c, d] \subseteq I \right\} < \infty.$$

Prop. If a function $f: [0, \infty) \rightarrow [0, \infty)$ is Riemann-Integrable on $[0, A]$ for all $A > 0$, then

$$\int_{[0, \infty)} f = \lim_{A \rightarrow \infty} \int_0^A f.$$

Proof. Suppose that the improper integral converges.

First, given $A > 0$, $\int_0^A f \leq \int_{[0, \infty)} f$ by definition. Since f has only non-negative,

so $\int_0^A f$ increases monotonically, so the limit $\lim_{A \rightarrow \infty} \int_0^A f$ exists, with $\leq \int_{[0, \infty)} f$.
 secondly, by the definition and property of supremum, given $\varepsilon > 0$, there is $[c, d] \subseteq [0, \infty)$

$$\text{such that } \int_{[0, \infty)} f - \varepsilon < \int_c^d f \leq \int_0^A f.$$

$$\text{So, for } A > d, \int_{[0, \infty)} f < \int_c^d f + \varepsilon \leq \int_0^A f + \varepsilon \leq \lim_{A \rightarrow \infty} \int_0^A f + \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, we obtain

$$\int_{[0, \infty)} f \leq \lim_{A \rightarrow \infty} \int_0^A f.$$

$$\text{Consequently, } \int_{[0, \infty)} f = \lim_{A \rightarrow \infty} \int_0^A f. \quad \square$$